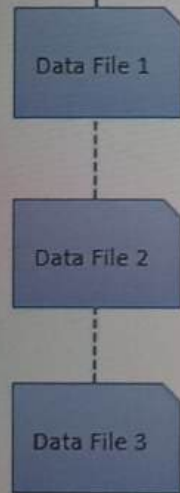
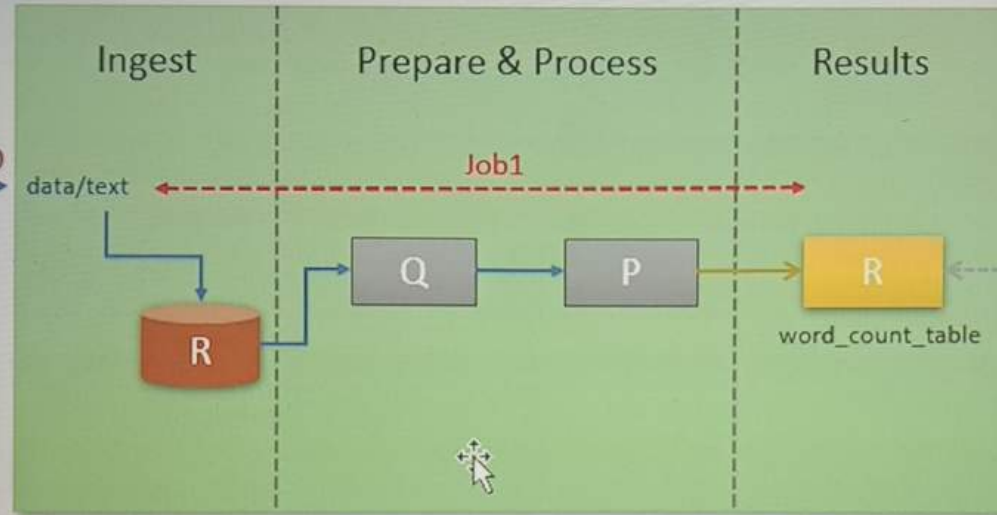


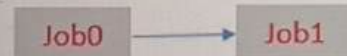
Source System



Data Engineering Platform



Data Pipeline (Workflow)



What is the frequency?
30 seconds

Challenges

1. Full Load -> Full Process -> **Overwrite Full Table**
2. Backpressure

Consumer Application



Word By Letters



01-streaming-word-count

Python

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```
class streamWC():
    def __init__(self):
        self.base_data_dir = "/FileStore/data_spark_streaming_scholarnest"

    def getRawData(self):
        from pyspark.sql.functions import explode, split
        lines = (spark.readStream
                  .format("text")
                  .option("lineSep", ".")
                  .load(f"{self.base_data_dir}/data/text")
                 )
        return lines.select(explode(split(lines.value, " ")).alias("word"))

    def getQualityData(self, rawDF):
        from pyspark.sql.functions import trim, lower
        return ( rawDF.select(lower(trim(rawDF.word)).alias("word"))
                 .where("word is not null")
                 .where("word rlike '[a-z]')")

    def getWordCount(self, qualityDF):
        return qualityDF.groupBy("word").count()
```



01-streaming-word-count Python ▾

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```
def getWordCount(self, qualityDF):
    return qualityDF.groupBy("word").count()

def overwriteWordCount(self, wordCountDF):
    return (wordCountDF.writeStream
            .format("delta")
            .option("checkpointLocation", f"{self.base_data_dir}/checkpoint/word_count")
            .outputMode("complete")
            .toTable("word_count_table")
            )

def wordCount(self):
    print(f"\tStarting Word Count Stream...", end="\n")
    rawDF = self.getRawData()
    qualityDF = self.getQualityData(rawDF)
    resultDF = self.getWordCount(qualityDF)
    sQuery = self.overwriteWordCount(resultDF)
    print("Done")
    return sQuery
```



Shift+Enter to run



Udemy



Python



```
class streamWCTestSuite():
    def __init__(self):
        self.base_data_dir = "/FileStore/data_spark_streaming_scholarnest"

    def cleanTests(self):
        print(f"Starting Cleanup...", end='')
        spark.sql("drop table if exists word_count_table")
        dbutils.fs.rm("/user/hive/warehouse/word_count_table", True)

        dbutils.fs.rm(f"{self.base_data_dir}/chekpoint", True)
        dbutils.fs.rm(f"{self.base_data_dir}/data/text", True)

        dbutils.fs.mkdirs(f"{self.base_data_dir}/data/text")
        print("Done\n")

    def ingestData(self, itr):
        print(f"\tStarting Ingestion...", end='')
        dbutils.fs.cp(f"{self.base_data_dir}/datasets/text/text_data_{itr}.t
base_data_dir}/data/text/")
        print("Done")
```



02-word-count-test-suite

Python ▾

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```
def runTests(self):
    import time
    sleepTime = 30

    self.cleanTests()
    wc = streamWC()
    wc.wordCount()

    print("Testing first iteration of batch word count...")
    self.ingestData(1)
    print(f"\tWaiting for {sleepTime} seconds...")
    time.sleep(sleepTime)
    self.assertResult(25)
    print("First iteration of batch word count completed.\n")

    print("Testing second iteration of batch word count...")
    self.ingestData(2)

    self.assertResult(32)
    print("Second iteration of batch word count completed.\n")

    print("Testing third iteration of batch word count...")
```



Udemy



```
def runTests(self):
    import time
    sleepTime = 30

    self.cleanTests()
    wc = streamWC()
    sQuery = wc.wordCount()

    print("Testing first iteration of batch word count...")
    self.ingestData(1)
    print(f"\tWaiting for {sleepTime} seconds...")
    time.sleep(sleepTime)
    self.assertResult(25)
    print("First iteration of batch word count completed.\n")

    print("Testing second iteration of batch word count...")
    self.ingestData(2)
    print(f"\tWaiting for {sleepTime} seconds...")
    time.sleep(sleepTime)
    self.assertResult(32)
    print("Second iteration of batch word count completed.\n")
```



02-word-count-test-suite

Python ▾

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```
print("Second iteration of batch word count completed.\n")

print("Testing third iteration of batch word count...")
self.ingestData(3)
print(f"\tWaiting for {sleepTime} seconds...")
time.sleep(sleepTime)
self.assertResult(37)
print("Third iteration of batch word count completed.\n")

sQuery.stop()
```

Shift+Enter to run

Shift+Ctrl+Enter to run selected text



02-word-count-test-suite

Python ▾

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Interrupt

● my cluster ▾

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Python



```
swcTS = streamWCTestSuite()  
swcTS.runTests()
```

*** Running command...

▶ (4) Spark Jobs

Starting Cleanup...Done

Starting Word Count Stream...Done

Testing first iteration of batch word count...

Starting Ingestion...Done

Waiting for 30 seconds...

Shift+Enter to run

Shift+Ctrl+Enter to run selected text



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